FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: thermal
Model status: BOUNDED_IEC_60287_CORE_EQUATION
Captured UTC: 2026-10-26T06:04:38.310Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: ambientC | Entered value: 40
Input path: conductorLimitC | Entered value: 90
Input path: racOhmPerM | Entered value: 0.00013
Input path: loadedConductors | Entered value: 3
Input path: t1KmPerW | Entered value: 0.23
Input path: t2KmPerW | Entered value: 0.08
Input path: t3KmPerW | Entered value: 0.06
Input path: t4KmPerW | Entered value: 0.76
Input path: dielectricLossWPerM | Entered value: 0.08
Input path: sheathLossFactor | Entered value: 0.17
Input path: armourLossFactor | Entered value: 0
Input path: loadCurrentA | Entered value: 225

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: BOUNDED_IEC_60287_CORE_EQUATION
Calculated output path: method | Value: IEC 60287-1-1 steady-state AC core rating equation using explicitly precomputed Rac, T1...T4, dielectric Wd and lambda loss factors. NOT a complete IEC 60287 design: no self-calculated thermal environment, thermal resistivity, AC skin/proximity, sheath circulating losses, trefoil grouping, solar drying, soil moisture, cable construction corrections or 60853 cyclic rating.
Calculated output path: ratedCurrentA | Value: 336.1222803507795
Calculated output path: continuousPerConductorA | Value: 336.1222803507795
Calculated output path: conductorTemperatureRiseAllowanceC | Value: 50
Calculated output path: dielectricTemperatureRiseC | Value: 0.22520000000000004
Calculated output path: conductorHeatingCoefficient | Value: 0.00044056999999999994
Calculated output path: enteredLoadCurrentA | Value: 225
Calculated output path: approxConductorCAtFixedRatedR | Value: 62.52905625
Calculated output path: loadingPercent | Value: 66.93992429338171
Calculated output path: withinEnteredCoreRating | Value: true
Calculated output path: missingForFullStandard[0] | Value: Temperature-adjusted AC resistance including skin/proximity
Calculated output path: missingForFullStandard[1] | Value: IEC 60287-2 installation thermal resistances
Calculated output path: missingForFullStandard[2] | Value: Metal sheath and armour loss factors
Calculated output path: missingForFullStandard[3] | Value: Parallel cable/grouping and mutual heating
Calculated output path: missingForFullStandard[4] | Value: Solar radiation and soil moisture
Calculated output path: missingForFullStandard[5] | Value: Manufacturer cable temperature/dielectric loss parameters
Calculated output path: missingForFullStandard[6] | Value: Validated operating power-flow current

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual

manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: thermal
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:38.310Z